

Tram Minh Man

(+65) 83937334 | man.tram@u.nus.edu | <https://www.linkedin.com/in/man-tram/> | <https://github.com/DanielJames0302>

[Tram Minh Man's Portfolio Website](#)

Education

Aug 2023 - Present	National University of Singapore Bachelor of Computing (Honours) in Computer Science	Singapore
Sep 2020 – May 2023	Le Hong Phong High School for The Gifted Informatics Honors Class, High School Diploma	Vietnam

Work Experience

Aug 2025 – Present	Health Informatics Research Lab <i>Software Engineer</i> - Architected and developed a real-time chat service using Next.js, TypeScript, and AWS Amplify GraphQL API, achieving sub-200ms message delivery latency through optimized GraphQL subscriptions. - Implemented real-time messaging with GraphQL subscriptions for instant message delivery and updates, processing 50+ messages per batch with cursor-based pagination. - Created reusable React hooks for chat management, message subscriptions, and search with proper state management, reducing component re-renders by 40% through memoization and optimized state updates.	Singapore
May 2025 – Jul 2025	Amaris.AI <i>AI Engineering Intern</i> - Developed AI report generation pipelines using LlamaIndex and LangGraph to aggregate, analyze, and summarize multi-source search results, enhancing knowledge extraction and delivery for end-users. - Developed and optimized API key management and rotation logic, ensuring high availability and uninterrupted access to third-party search/data services. - Built and containerized an Agent-to-Agent backend system with Docker, enabling secure, efficient, and automated data exchange between multiple microservices.	Singapore

Mar 2025 - May 2025

Smart Systems Institute
Research Intern

Singapore

- Manually annotated road images to fine-tune a pre-trained SAM model on a dataset of over 15,000 images, achieving segmentation accuracy above 90%.
- Utilized a state-of-art calibration method to calibrate robot's camera, significantly reducing error in converting between world coordinates and image coordinates.
- Designed an effective cost function and applied k-means algorithm to select most optimal candidate trajectory for robot, significantly eliminating unnecessary or dangerous paths.

Sep 2024 – Jan 2025

GreatFrontEnd
Software Engineering Intern

Singapore

- Developed key features and resolved critical technical bugs in the company's coding interview platform product, enhancing stability and performance with Next.js and TailwindCSS.
- Collaborated with marketing team to design technical and promotional content, contributing to a growth of 3,000 subscribers across company's social media accounts.
- Verified and authored detailed solution explanations for 100+ data structures and algorithms (DSA) questions, improving user learning experience.

Projects

Oct 2025 – Nov 2025

Chinese-English Machine Translation

Singapore

- Developed end-to-end training pipeline for Chinese-English machine translation using mT5 transformer models (300M-1.2B parameters) with PyTorch and HuggingFace Transformers.
- Implemented LoRA (Low-Rank Adaptation) fine-tuning to reduce memory consumption by 70%+ while maintaining translation quality, enabling training of large models on limited GPU resources.
- Built multi-GPU distributed training system with DDP support and memory-optimized evaluation pipeline, reducing training time by 3x through gradient accumulation and efficient logit preprocessing.

Sep 2025 - Dec 2025

CAPTCHA Solver

Singapore

- Built end-to-end CRNN training pipeline with TensorFlow/Keras, implementing residual CNN + BiLSTM architecture, custom CTC loss, and automated data augmentation for CAPTCHA text recognition
- Developed and trained DCGAN models in PyTorch for synthetic CAPTCHA generation, utilizing spectral normalization and hinge loss for stable adversarial training on grayscale and RGB character datasets
- Designed modular ML infrastructure with custom data providers, evaluation metrics (CER/WER),

Sep 2025 - Dec 2025

Peerprep

	- Developed a collaborative coding practice platform with Next.js implementing authentication flows, real-time matching interface, and collaborative coding practice rooms. - Implemented high-performance matching algorithm using Redis pipelines and distributed locking, processing matches every 1 second with 60% latency reduction and supporting 100+ concurrent WebSocket connections with sub-100ms response times. - Deployed 6 microservices to GCP Cloud Run with auto-scaling (1-5 instances) and optimized resource allocation, reducing infrastructure costs by 35% while maintaining 99.95% uptime with load balancing across 6 backend services.	
Jul 2024 - Sep 2024	Gestura - Developed an AI-powered social app with Next.js to detect sign language, significantly enhancing accessibility and user experience for the deaf-mute community. - Finetuned MediaPipe gesture recognizer model to accurately detect and interpret sign language gestures in videos, resulting in an accuracy of 40%. - Built a backend system with FastAPI, developing RESTful APIs to manage video uploads, processing, and retrieval of AI detection results.	Singapore
May 2024 - Jul 2024	Foodie - Built robust backend functionality with PostgreSQL, and Go, including user authentication, post management, follow requests, and file uploads utilizing RESTful APIs and JWT (JSON Web Tokens) for secure authentication. - Designed and structured backend architecture with modular components such as controllers, models, routes, and middleware for efficient development and maintenance. - Utilized Pusher's Web Sockets to facilitate instant message delivery for effective real-time communication between multiple users.	Singapore

Scholastic Achievements/Extracurricular Activities

May 2025	Earth Rover Challenge (ERC) at IEEE International Conference on Robotics & Automation 2025 First Prize with project "GeNIE: A Generalizable Navigation System for In-the-Wild Environments"	Singapore
Jan 2024 - May 2024	School Contributions Teaching Assistant for CS1010 Programming Methodology course at NUS School of Computing	Singapore
Mar 2023	Ho Chi Minh City Science and Engineering Fair First Prize in the Intelligent Systems Category with project "Smart Eyewear Device for Deaf-mute Community"	Vietnam

Skill Sets

Programming	Python Go Javascript Typescript	
--------------------	--	--

	SQL
Web	HTML, CSS
Database	PostgreSQL Google Firebase
Web Frameworks and Libraries	FastAPI React Next.js Tailwind CSS
AI/ML Frameworks and Libraries	PyTorch Numpy Pandas LangChain LangGraph LlamaIndex Matplotlib OpenCV
DevOps Tools	Docker
Tools	Git Github
Office Productivity	Microsoft Word, PowerPoint Microsoft Excel
Non-technical Skills	Outlining technical content blogs/posts Video editing

Language Proficiency

Spoken	English – fluent;
Written	English – fluent;